

Generating and Comparing Two Patterns

Answer Key

Grades 4–5 Mathematics

Quick Answers

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|---|---|
| 1. 20 | 7. 24 |
| 2. 40 | 8. x-coordinate: pattern Q term 5 = 18, y-coordinate: pattern P term 5 = 19 |
| 3. 2 | 9. (a) mean of pattern P = 13, (b) mean of pattern Q = 10 |
| 4. 36 | 10. (a) 12th term of pattern A = 44, (b) 12th term of pattern B = 88, — |
| 5. x-coordinate: pattern A term 4 = 12, y-coordinate: pattern B term 4 = 24 | |
| 6. 4 weeks | |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

5.G.A.1–A.2 – *problems 5, 8*

5.OA.B.3 – *problems 1, 2, 3, 4, 6, 7, 9, 10*

Difficulty 1–5 of 5 across 15 tagged checks.

- 1.** Write the 6th term of pattern A (start at 0, add 4).

Solution: The 6th term is 5 steps past the start, not 6 — the first term is the start itself. So $0 + 5 \times 4$. Listing confirms it: 0, 4, 8, 12, 16, 20. 20

- 2.** Write the 6th term of pattern B (start at 0, add 8).

Solution: Same count of steps, larger step: $0 + 5 \times 8$. Listing: 0, 8, 16, 24, 32, 40. 40

- 3.** Each time pattern A adds its step, pattern B adds a step how many times as large?

Solution: Compare the steps themselves, not the terms: pattern B adds 8 where pattern A adds 4, so divide $8 \div 4$. 2

This single fact is what makes every later comparison on this sheet come out the way it does.

4. How much greater is the 10th term of pattern B than the 10th term of pattern A?

Solution: Both 10th terms are 9 steps past a start of 0. Pattern B: $0 + 9 \times 8 = 72$. Pattern A: $0 + 9 \times 4 = 36$. Subtract: $72 - 36$. 36

5. Write the ordered pair (pattern A's 4th term, pattern B's 4th term).

Solution: Both 4th terms are 3 steps past 0. Pattern A: $0 + 3 \times 4 = 12$, which is the across number. Pattern B: $0 + 3 \times 8 = 24$, the up number. (12, 24)

6. Maya has 12 dollars and adds 3 dollars a week; Liam has 4 dollars and adds 5 dollars a week. After how many weeks are they equal?

Solution: Generate both patterns week by week.

Maya: 12, 15, 18, 21, 24. Liam: 4, 9, 14, 19, 24.

They first agree in the fifth entry of each list, which is after 4 weeks of saving. Algebraically, $12 + 3w = 4 + 5w$ gives $8 = 2w$. 4 weeks

Liam starts 8 dollars behind and gains 2 dollars a week on Maya, so he needs $8 \div 2$ weeks to catch up.

7. How many dollars will each have at the moment their savings are equal?

Solution: Put the number of weeks back into either rule — both must give the same total, which is the check. Maya: $12 + 3 \times 4$. Liam: $4 + 5 \times 4 = 24$ as well. 24

8. From the table of patterns P and Q, write the ordered pair (Q's 5th term, P's 5th term).

Solution: Read down the Q row to term 5, which is the across number: 18. Then read the P row to term 5 for the up number: 19. The order named in the question decides which goes first. (18, 19)

9. (a) Mean of pattern P's five terms. (b) Mean of pattern Q's five terms.

Solution (a): $7 + 10 + 13 + 16 + 19 = 65$, and $65 \div 5$. 13

(b) $2 + 6 + 10 + 14 + 18 = 50$, and $50 \div 5$. 10

Both means land exactly on the middle term of their own pattern. That is not luck: in a pattern that grows by a fixed step, the terms are balanced around the middle one.

10. Patterns A and B again. (a) 12th term of A. (b) 12th term of B. (c) Describe and justify the relationship between matching terms.

Solution (a): 11 steps past 0: $0 + 11 \times 4$. 44

(b) 11 steps past 0 with the larger step: $0 + 11 \times 8$. 88

(c) — **instructor-judged.** The relationship is that every term of pattern B is double the matching term of pattern A.

Look for a reason, not only the pattern spotted: both patterns start at the same place, 0, and pattern B adds twice as much on every single step, so after any equal number of steps B has added exactly twice as much as A. A response listing three matching pairs and stopping has noticed the rule but not shown why it cannot break.